

Intelligent manufacturing and enterprise low-carbon transformation: Evidence from Chinese industrial firms

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Abstract: The rapid growth of artificial intelligence technology has made intelligent manufacturing the mainstream trend. Whether artificial intelligence technology helps solve global warming issues and improve the environment has garnered widespread attention. In this study, we examine whether artificial intelligence technologies, represented by industrial robot adoption, contribute to the low-carbon transformation of manufacturing enterprises. Utilizing panel data of Chinese manufacturing enterprises from 2007 to 2016, we find that the adoption of robots is associated with a notable reduction in the carbon emission intensity of enterprises, productivity enhancement, labor cost saving and low-carbon technological innovation are the main channels. Further analysis indicates that the adoption of robots prompts enterprises with low-carbon patents to adjust their production range, thereby reducing the structural intensity of carbon emissions, and the reduction in carbon emissions from manufacturing enterprises has not led to carbon leakage. In conclusion, this study provides micro-level evidence for intelligent manufacturing facilitating the low-carbon transformation of enterprises. It also offers guidance for countries on how to better achieve carbon neutrality goals.

Keywords: Intelligent manufacturing; Robot adoption; Carbon emission intensity; Low-carbon transformation

1.Introduction

The rise of artificial intelligence is hailed as Industry 4.0, bringing profound and widespread impacts to society (Beier et al., 2020; Beier et al., 2022). The adoption of industrial robots, as a crucial component of artificial intelligence, plays a significant supporting role in various stages of production and manufacturing (Liu et al., 2021; Li et al., 2022), which could profoundly affect energy use (Huang et al., 2022) and carbon

emissions closely related to firms' industrial production. Meanwhile, the carbon emissions generated by enterprise production have led to severe climate issues, imposing significant negative externalities on the global environment (Li et al., 2022). In this context, exploring whether intelligent manufacturing contributes to the low-carbon transformation of manufacturing enterprises and its impact mechanism can help us further understand the relationship between technological advancement and climate change. Additionally, it provides theoretical basis and policy guidance for countries on how to achieve carbon neutrality goals, holding both theoretical and practical significance.

As a manufacturing powerhouse, China leads the world in robot adoption and total carbon emissions. On the one hand, due to the outstanding performance of robots in reducing labor costs and enhancing production efficiency, China's government has identified the robotics industry as a strategically important sector (Cheng et al., 2019). This has led to a rapid growth in the adoption of industrial robots in China. According to statistics from the International Federation of Robotics (IFR), China is currently the world's largest consumer market for industrial robots. In 2022, the new installations reached 290,258 units, surpassing the combined total of new installations in other countries worldwide. With a growth rate of 5%, this figure exceeded the previous record set in 2021, signaling China's ongoing transformation towards intelligent manufacturing. On the other hand, as a major manufacturing nation, China's production has resulted in substantial energy consumption and carbon emissions (Lin and Xu, 2019). According to the International Energy Agency's (IEA) "CO₂ Emissions in 2022" report, China's total carbon emissions reached 12.1 billion tons in 2022, accounting for nearly one-third of the global total of 36.8 billion tons. As a responsible major nation, China explicitly sets forth dual carbon goals in September 2020: achieving carbon peaking goals by 2030 and carbon neutrality goals by 2060 (Yu et al., 2023; Zhang and Zhu, 2023). In pursuit of emission reduction, Chinese government has initiated a nationwide carbon emissions trading market and outlined specific provisions in the 14th Five-Year Plan to reduce energy intensity by 13.5% compared to the levels in 2020 by the year 2025. China is making tremendous efforts to reduce its own carbon emissions.

Considering the comprehensive factors mentioned above, conducting research on whether intelligent manufacturing contributes to the low-carbon transformation of enterprises using Chinese enterprises as research samples not only yields more persuasive conclusions but also helps more developing countries draw lessons from China's experience to achieve carbon neutrality goals.

To better elucidate how industrial robot adoption influences carbon emissions at the micro-level of enterprises, we employed energy consumption data from the Annual Tax Survey (ATS) for Chinese manufacturing enterprises from 2007-2016 to calculate the carbon emissions per unit of output, i.e., carbon emission intensity. Furthermore, we integrated information on industrial robot imports from the China Customs Database (CCD) to investigate how robot adoption affects the carbon emission intensity of enterprises. The research findings indicate that the adoption of industrial robots significantly reduces the carbon emission intensity of enterprises, thereby promoting their transition towards a low-carbon footprint. Based on the results, this paper discusses the reasons for the promotion of low-carbon transformation by robot adoption. Firstly, the adoption of industrial robots enhances the overall factor productivity of enterprises, leading to a reduction in energy consumption during the production process and a decrease in carbon emission intensity due to improved production efficiency. Secondly, the adoption of industrial robots can save on labor costs, and these cost savings provide enterprises with additional funds for carbon reduction-related activities, thereby lowering the carbon emission intensity of the enterprises. Lastly, the adoption of industrial robots stimulates enterprises to engage in the research and development of low-carbon technologies, manifested by an increase in the number of green patents held by the enterprises, thus fostering the low-carbon transformation and serving as a crucial factor in reducing carbon emission intensity. Subsequently, the paper further discusses how the adoption of industrial robots affects the overall carbon emissions of enterprises. The research reveals that the extensive use of industrial robots significantly increases the output of enterprises. Although the carbon emission intensity of enterprises decreases, the overall carbon emissions still show an increase due to the substantial growth in output. Also, this paper conducts a detailed analysis of whether robot

adoption influences the range of products manufactured by enterprises, and we find that the enterprises with more low-carbon patents tend to reduce the carbon emission structural intensity of exported products after the adoption of robots, thereby promoting the low-carbon transformation of enterprises. Finally, through regression analysis on the intensity of electricity usage in the manufacturing sector, we found that the decarbonization effect of robots does not result in carbon leakage to other sectors.

Specifically, the potential contributions of this paper may be summarized as follows: Firstly, to the best of our knowledge, this paper represents the first attempt to analyze how the adoption of industrial robots affects the carbon emission intensity of manufacturing enterprises using large-scale micro-level enterprise data. Existing research on the relationship between industrial robot adoption and carbon emissions has primarily focused on more macro-level industry or city perspectives, while micro-level studies have concentrated on the impact of robot adoption on employment, innovation, or energy efficiency. This paper directly investigates whether the adoption of industrial robots can reduce the carbon emission intensity of enterprises, providing a valuable complement to existing research outcomes. Secondly, this paper conducts a micro-level analysis of how micro-level entities utilize artificial intelligence technologies, represented by robots, to reduce carbon emission intensity through channels such as increased productivity, labor cost savings, and greater innovation in low-carbon technologies. Micro-level enterprises are the main users of robots and the sources of carbon emissions, and a mechanism analysis at the enterprise level can better reveal how robot adoption lowers carbon emission intensity, thereby enhancing the persuasiveness of the conclusions. Thirdly, we further explore how robot adoption affects the range of goods produced by enterprises and whether its emission reduction effect leads to carbon leakage. We find that enterprises with more low-carbon patents experience a significant reduction in carbon emission structural intensity after adopting robots. Additionally, robot adoption does not lead to an increase in enterprise electricity consumption, thus averting carbon leakage. This broader perspective provides richer evidence for how industrial robots promote the low-carbon transformation of enterprises.

The remainder of this paper proceeds as follows. Section 2 presents the literature review. Section 3 introduces theoretical hypotheses. Section 4 presents the data and empirical method. Section 5 presents the empirical results and robustness checks. Section 6 analyzes the mechanisms and conducts heterogeneity analysis. Section 7 presents additional analyses. Section 8 summarizes the conclusions and presents some policy implications.

2. Literature review

The literature relevant to this paper can be categorized into three branches: one focusing on the influencing factors of carbon emissions, another on the economic consequences of industrial robots, and the final one on how industrial robots impact carbon emission intensity.

2.1 Factors influencing carbon emissions

At the macro level, policy design significantly impacts carbon emissions. For instance, carbon emissions trading markets play a crucial role as a decarbonization tool. The European Union Emissions Trading System (EU-ETS) stands out as the earliest-established global carbon trading market, initiated in 2005 and currently in its fourth compliance period. Guo et al. (2020) found that the profit obtained by enterprises is positively correlated with the reduction in carbon emissions when utilizing carbon trading mechanisms, and the reduction in national carbon emissions is primarily attributed to contributions from industrial giants. Bayer and Aklin (2020) confirmed the substantial reduction in EU carbon emissions attributed to EU-ETS, despite the relatively low carbon prices in the market. With the initiation of China's carbon emissions trading pilot program in 2013 and the nationwide launch of the carbon emissions trading market in 2017, increasing attention has been directed toward the emission reduction effects of China's carbon trading market. Utilizing empirical research methods, Zhang and Wang (2024) discovered that the carbon market significantly enhances the energy efficiency of power enterprises. Research by Lin and Jia (2019) and Yu et al. (2023), employing CGE simulations, demonstrated the carbon market's favorable impact on emission reduction.

Beyond carbon markets, other factors also influence carbon emissions. Tang et al. (2024) found that increased pressure on official promotions leads to a greater emphasis on economic growth, consequently raising enterprise carbon emissions. Xing et al. (2024), in a city-level study, indicated that energy usage, total city output, urban area, and city population are crucial factors affecting urban carbon emissions. Concurrently, research by Li and Zhang (2023) and Ding and Liu (2024) revealed that technological progress serves as a significant driving force in reducing carbon emission intensity. Altunbas et al. (2023) discovered that acquisition activities temporarily increase enterprise carbon emission intensity, with the acquiring party's production efficiency not necessarily translating into innovation in low-carbon technologies. However, Jiang et al. (2023), focusing on the supply chain perspective, arrived at a different conclusion, revealing that low-carbon behaviors among upstream and downstream partners induce a ripple effect, promoting the low-carbon transformation of the entire supply chain through learning and competitive effects.

2.2 Economic consequences of industrial robot adoption

Another branch of research relevant to this paper explores the economic consequences of industrial robot adoption. Firstly, according to the IFR definition, industrial robots are programmable, multipurpose machines used for automatic control operations in industrial automation. The application of industrial robots can assist humans in completing repetitive tasks, thereby impacting the labor market. Representative literature in this area includes Acemoglu and Restrepo (2020), who, building upon Zeira's (1998) task model, specifically describe the substitution process of artificial intelligence for labor. The theoretical framework they propose serves as a paradigm for future research in the field of robotics. Subsequently, utilizing IFR data to construct a robot exposure index, they found that robot adoption lowered employment and wage levels in the United States. Another study by Acemoglu et al. (2020), based on French company-level data, found that industrial robot adoption decreases labor share. Koch et al. (2021), using Spanish company data, reached similar conclusions, further confirming the substitutive effect of robots on labor. Additionally, Graetz and Michaels (2018) demonstrated that the substitution effect of industrial robot

adoption is more pronounced among low-skilled workers.

In addition, existing research examines whether industrial robot adoption contributes to achieving economic growth objectives. Differing from Acemoglu and Restrepo (2020), Aghion (2017) introduces the concept of "cost disease" into Zeira's (1998) model, finding that slow-growing sectors become increasingly important in the economy, and robots, automation, and artificial intelligence can alleviate this contradiction to some extent. Graetz and Michaels (2018) conducted empirical research using national panels and found that emerging production methods like industrial robots increased the rate of economic growth by 0.37%. Regarding other economic consequences of industrial robot adoption, on the one hand, the application of artificial intelligence technologies can promote enterprise innovation (Aghion et al., 2017). Liu et al.'s (2020) research confirms this viewpoint and suggests that channels such as knowledge creation and spillover, learning and absorption, and R&D investment are crucial for promoting innovation. On the other hand, robot adoption can enhance enterprise productivity (Graetz and Michaels, 2018; Ballestar et al., 2020), with Duan et al. (2023) pointing out that, apart from innovation, optimizing human capital structure is also a significant mechanism for improving productivity.

As we concentrate on the economic impact of robot adoption in the Chinese context, we further review the evidence provided from China in some studies. Similar to studies focusing on other countries, research on China also addresses how robot adoption affects employment. However, conclusions vary due to the utilization of different samples and identification methods. Wang and Dong (2020), utilizing data from listed companies and IFR, constructed a firm-level indicator of robot exposure using labor quantity as a weight. They found that robot adoption reduces firms' demand for labor, but the impact on wages is not significant. On the other hand, Li et al. (2022), merging data from Chinese industrial enterprises with Chinese Customs Database, identified robot adoption through robot imports. They found that it increases firms' demand for labor. In addition to its impact on employment, evidence from China suggests that robot adoption can effectively promote innovation (Feng et al., 2023). However, Zhu et al. (2022) found the opposite conclusion, suggesting that robot

adoption inhibits innovation, with technological selection effects and industry competition effects being the main channels. Similarly, some studies take an international perspective. Lyu et al. (2020) found that robot adoption benefits Chinese enterprises' participation in global value chain division of labor. Liu et al. (2020) further discovered that robot adoption enhances China's industry participation and position in the global value chain. He et al. (2024) found that industrial robot adoption significantly enhances the resilience of manufacturing enterprises in industrial chain supply chains.

2.3 Industrial robots and enterprise low-carbon transformation

In recent years, an increasing number of studies have focused on the energy and environmental effects of industrial robots, specifically whether they can improve energy efficiency, further reduce carbon emissions, alleviate the environmental burden of economic development, and drive the entire economy towards low-carbon transformation. Some studies conducted at the national level include Li et al. (2022), who utilized panel data from 35 countries spanning 1993 to 2017 and found that industrial robot adoption can reduce the overall carbon intensity of a country. However, Zhang and Zhu (2023) reached the opposite conclusion, suggesting that industrial robot adoption increases a country's overall carbon emissions. Zhao et al. (2024) found that the adoption of artificial intelligence technology can mitigate internal carbon emission inequality within a country. At the industry level, Wang et al. (2022) used a cross-national industry panel and the GMM method to discover that industrial robots significantly improve energy efficiency in manufacturing. Similarly, Liu et al. (2021) and Liu et al. (2022) found similar conclusions at the industry level. More research has been conducted at the city level, such as those by Yu et al. (2023) and Liu et al. (2024), discovered that industrial robot adoption can reduce carbon emissions. However, Li et al.'s (2024) research found an "N"-shaped relationship between the digital economy and carbon emission efficiency, with the threshold variable being green technology innovation. Li et al.'s (2023) study revealed that the implicit carbon emissions of the digital economy are more significant than direct carbon emissions, mainly due to intermediate inputs from non-digital sectors.

Existing research has mainly been conducted at the relatively macro levels of countries, industries, or cities, with a relative lack of studies at the micro-level of enterprises. However, micro-level enterprises constitute the primary users of industrial robots and the main sources of carbon emissions. Research conducted at the city level, industry level or national level is the sum of firm behavior at the macro level. Analyzing at the enterprise level can better reveal the channels through which robot adoption reduces carbon emissions. Unfortunately, the current academic literature on how robot adoption at the micro-level affects carbon emissions is not yet sufficient, only Zhou and Liu (2024) adopted the data of 10 listed manufacturing companies in China from 2013 to 2022 to prove that the digital transformation of enterprises significantly reduces the carbon emissions of enterprises, the carbon reduction effect is more pronounced in state-owned enterprises, and digital finance can enhance the carbon reduction effect of digital transformation. However, due to the small number of samples, the conclusions drawn are not representative.

Certainly, there is some research relevant to the theme of intelligent manufacturing and enterprise low-carbon transformation. Existing research conducted at the micro-level suggests that the application of artificial intelligence technologies can stimulate enterprise innovation (Aghion et al., 2017) and enhance productivity (Ballestar et al., 2020; Duan et al., 2023). Technological advancements further facilitate the green transformation of enterprises (Shapiro and Walker, 2018), leading to improved energy efficiency (Huang et al., 2022), and more green innovation outcomes (Tang et al., 2023; Gan et al., 2023). These aspects are considered crucial features of the impact of artificial intelligence on micro-level enterprises. In this context, the study conducted by Huang et al. (2022) shares similarities with the present research. The focus of their work lies in examining the impact of robot adoption on enterprise energy performance. They discovered that the adoption of robots significantly reduces the energy intensity of enterprises by enhancing total factor productivity. Compared to Huang et al. (2022), our study focuses on enterprise carbon emissions. In addition to the productivity effect, we found that cost savings and low-carbon innovation are crucial channels through which the adoption of robots influences the carbon emission intensity of enterprises.

Furthermore, our research provides more comprehensive evidence from the perspective of adjusting the range of products manufactured by enterprises and the reduction in carbon emission intensity from manufacturing enterprises has not led to carbon leakage.

3. Theoretical hypotheses

3.1 Productivity enhancement effect

Industrial robots can achieve uninterrupted production 24 hours a day, executing repetitive and tedious tasks at higher speeds and consistent quality. Due to the ability of robots to work continuously without fatigue or human limitations, they can achieve continuous production, reducing downtime in production. Moreover, they execute tasks more accurately and quickly, contributing to the overall improvement of enterprise productivity. Currently, the consensus in existing research is that the adoption of robots can enhance productivity (Graetz and Michaels, 2018; Ballestar et al., 2020; Duan et al., 2023). The increased productivity allows enterprises to utilize resources more efficiently, including energy and raw materials. As a result, enterprises can achieve outputs at levels equivalent to previous ones with less energy input during the production process, thereby lowering carbon emission intensity. Additionally, increased productivity may come with more effective waste disposal and emission management. Through more efficient production processes and stricter waste management, enterprises can reduce the generation of harmful waste and emissions, thereby decreasing environmental pollution and carbon emissions. Therefore, the improvement in productivity contributes to enhancing the efficiency of energy usage in enterprises, reducing carbon emission intensity, and facilitating the transition to a low-carbon economy. Based on this, we propose:

Hypothesis 1: The adoption of industrial robots, through the productivity enhancement effect, reduces the carbon emission intensity of enterprises.

3.2 Labor cost saving effect

Firstly, industrial robots can replace humans in performing a substantial amount of repetitive work. Secondly, robots are often programmable and reconfigurable based on specific tasks, making them adaptable to different production demands and

facilitating customization and personalized production to meet consumer needs. This flexibility enables businesses to respond more effectively to market changes without the need for extensive manpower adjustments. Finally, in hazardous or harmful environments, the adoption of robots can reduce human risks, decreasing the likelihood of personnel exposure to dangerous conditions. This contributes to lowering costs associated with employee health and safety.

In summary, the adoption of robots can substitute for a significant portion of non-technical labor, resulting in labor cost savings. Upon achieving cost savings in labor through the adoption of industrial robots, businesses can allocate the saved funds to other operational activities. In response to government calls for achieving dual carbon goals, enterprises may use the saved funds to invest in more clean energy or environmentally friendly equipment, reducing carbon emissions at the source. Alternatively, they may invest in carbon offset projects, such as tree planting and carbon capture technologies, to offset their carbon emissions. Both outcomes contribute to a reduction in the carbon emission intensity of the enterprise. Based on this, we propose:

Hypothesis 2: The adoption of industrial robots, through labor cost saving effect, reduces the carbon emission intensity of enterprises.

3.3 Low-carbon technological innovation effect

Firstly, industrial robots belong to advanced production technologies and automation systems. The adoption of these innovative technologies in the production process may involve novel and unique methods and systems, creating low-carbon technological innovations that reduce the carbon emission intensity of enterprises. Secondly, the substitution of low-skilled labor by robots promotes an upgrade in the human capital structure, leading to the employment of more high-skilled labor. According to human capital theory and endogenous economic growth theory, high-skilled labor is more inclined towards innovative research and development activities, increasing the enterprise's R&D expenditures and promoting progress in low-carbon technologies (Gan et al., 2023), thereby reducing the carbon emission intensity of the enterprise. Thirdly, industrial robots are typically designed as efficient, precise, and low-energy-consumption devices. By using these robots, enterprises can reduce energy

consumption (Liu et al., 2021; Huang et al., 2022), minimize waste generation, and adopt lower-carbon methods in the production process. This improvement in energy efficiency provides a foundation for enterprises to pursue low-carbon technological innovation, ultimately lowering the carbon emission intensity of the enterprise. Finally, robots, through sensors and data collection technologies, can comprehensively preserve data from the production process. This provides enterprises with more comprehensive and real-time information. By analyzing this data, enterprises can more easily identify inefficient aspects of production, propose improvement methods, enhance energy efficiency, promote innovation in low-carbon technologies, and reduce carbon emission intensity. Based on this, we propose:

Hypothesis 3: The adoption of industrial robots, through low-carbon technological innovation effect, reduces the carbon emission intensity of enterprises.

4. Data and empirical method

4.1 Introduction of data

The databases utilized in this study primarily consist of two datasets: the first set comprises data from the Annual Tax Survey (ATS), while the second set is derived from the Chinese Customs Database (CCD). These datasets can be matched based on the company name and year. The sample years for the former range from 2007 to 2016, while the latter spans from 2000 to 2016. To ensure uniformity in data years, we construct a panel of enterprises covering the period from 2007 to 2016 for analyzing how the adoption of robots affects enterprise carbon emission intensity. Subsequently, a detailed introduction of the databases is presented.

ATS is obtained through a sampling survey of Chinese enterprises, capturing various aspects of business operations, including the tax situation. However, the investigation into enterprises is not continuous, as sampled companies may not be surveyed every year. The data includes approximately 700,000 companies nationwide each year, encompassing key enterprises in various industries, large and medium-sized enterprises, as well as small and medium-sized enterprises, representing all types of businesses with high representativeness.

For the empirical study, we use a sample of manufacturing enterprises from the ATS. ATS provides basic information, including enterprise name, industry affiliation, and establishment year. Additionally, it offers financial statements such as profit and loss statements, balance sheets, and cash flow statements to reflect the enterprise's production and operational status. Furthermore, it includes information on the enterprise's energy use, aiding in the calculation of carbon emissions during the production process. The choice of analyzing the sample of manufacturing enterprises stems from their status as primary users of industrial robots and their relatively high contribution to total carbon emissions in China. According to the Carbon Emission Accounts & Datasets (CEADS), China's manufacturing sector contributes to over one-third of the country's total carbon emissions, with the power and heat production industries being the primary sources. Lowering carbon emissions in the manufacturing industry is considered a crucial means to achieve the carbon reduction objectives.

<Figure1 is inserted about here>

Import and export data are sourced from the CCD, which meticulously records the import and export information of various enterprises. This information includes the names of import and export enterprises, import (export) years, countries of import (export), HS8 codes of imported (exported) goods, and the value of imported (exported) goods. Leveraging customs data, we can not only identify which enterprises imported industrial robots in specific years but also measure changes in their production range by adjusting for changes in the export of goods.

4.2 Carbon emission intensity

The existing databases do not directly provide specific values for enterprise carbon emissions. Therefore, we resort to an indirect approach using the energy consumption of enterprises for calculation. The specific method is as follows: we multiply the energy consumption of enterprises in a given year by the corresponding emission factor of the respective energy source to obtain the total carbon emissions generated by the enterprise's production process that year (Barrows and Ollivier, 2021). ATS provides information on the consumption of coal and oil by enterprises in a given year, and the emission factors for these energy sources are referenced from the Intergovernmental

Panel on Climate Change (IPCC) "2006 IPCC Guidelines for National Greenhouse Gas Inventories". The carbon emission factors for coal and oil are 1.83 kilograms of carbon dioxide per kilogram and 3.07 kilograms of carbon dioxide per kilogram, respectively. After calculating the total carbon emissions of the enterprise using the above method, we divide it by the total output of the enterprise to obtain the carbon emission intensity per unit output, also known as carbon emission intensity.

The underlying logic of this calculation method assumes that, regardless of the production technology used, the carbon emissions generated in the process of using a given energy source are the same. While this assumption may appear somewhat arbitrary, it is considered reasonable in developing countries, especially in China within the time range of 2007 to 2016, where the dual carbon goals had not been proposed. During this relatively early period, enterprises were more focused on their own production and operational performance indicators, and environmental concerns, particularly carbon emissions, were relatively low. Only a few enterprises were willing to incur costs to adopt carbon capture technologies to reduce carbon emissions.

4.3 Robot adoption

In assessing the utilization of robots, we adopt the method proposed by Fan et al. (2021), and Huang et al. (2022). Following their approach, we use the information provided by the CCD to determine whether an enterprise imports industrial robots as an indicator of its deployment of robots in production. On one hand, the IFR data measures the overall level of robot adoption at the industry level without specifying which companies employ industrial robots in their production processes. On the other hand, according to IFR, in the early years, China did not possess the capability to produce industrial robots domestically, with over 70% of robot adoption being imported from abroad (Huang et al., 2022).

The specific measurement of robot adoption at the enterprise level is as follows: by merging ATS with CCD based on the enterprise name and year, we can determine whether each enterprise has imported industrial robots. If enterprise i imports industrial robots in year m , the variable $robot_{it}$ for enterprise i takes the value of 1 in year m and subsequent years, and 0 otherwise. To ensure that our research conclusions are not

influenced by the method of robot identification, we also provide regression results in robustness tests using industry-level robot stock data provided by the IFR database for re-identification. The conclusions remain robust in this alternate approach.

4.4 Empirical method

The following model (1) is employed in this study to examine the impact of industrial robot adoption on enterprise carbon emission intensity, where subscripts i , t and s represent the enterprise, year, and industry, respectively.

$$\ln ci_{it} = \beta_0 + \beta_1 robot_{it} + \lambda C_{it} + \varphi_i + \gamma_{st} + \varepsilon_{it} \quad (1)$$

The dependent variable $\ln ci_{it}$ represents the logarithmically transformed carbon emission intensity of enterprise i in year t , while the key independent variable $robot_{it}$ indicates whether enterprise i adopted robots in production in year t . Additionally, we introduce other variables at the enterprise level (C_{it}) that may affect carbon emission intensity, specified as follows: (1) Enterprise size ($\ln size_{it}$): measured by the logarithm of the average number of employees throughout the year; (2) Capital intensity of the enterprise ($\ln klratio_{it}$): measured by the logarithm of the total fixed assets of the enterprise divided by the number of employees; (3) Human capital level of the enterprise ($\ln hc_{it}$): measured by the logarithm of the average wage of the enterprise employees; (4) Profitability of the enterprise ($\ln profit_{it}$): measured by the logarithm of the net profit of the enterprise for the current year; (5) Type of the enterprise ($state_{it}$): determined by the registration type of the enterprise, with 1 indicating state-owned and 0 otherwise. Additionally, we introduce enterprise fixed effects φ_i in the regression to capture unobserved factors at the enterprise level that do not vary over time. Furthermore, we incorporate industry-time fixed effects γ_{st} to capture time-varying factors at the industry-year level affecting the carbon emission intensity of enterprises. Descriptive statistics for these variables are presented in Table 1.

In our baseline regression sample, there are a total of 618,574 manufacturing enterprises, with 1,390,554 observations. Among them, 958 enterprises utilize industrial robots, accounting for only 0.15% of the total, a figure consistent with the findings of Fan et al. (2021) and Huang et al. (2022).

Table 1: Statistical analysis results

Variables	Definition	Observations	Mean	S.D.
$ln ci_{it}$	ln(carbon intensity)	1390554	0.2558	0.6669
$robot_{it}$	whether the enterprise adopts industrial robots	1390554	0.0020	0.0442
$ln size_{it}$	ln(average number of employees per year)	1390554	4.1950	1.4187
$ln klratio_{it}$	ln(capital labour ratio)	1390554	3.8739	1.7014
$ln hc_{it}$	ln(per capita wage)	1390554	3.0602	0.8218
$ln profit_{it}$	ln(net profit)	1390554	1.7877	6.3026
$state_{it}$	whether the enterprise is a state-owned enterprise	1390554	0.0482	0.2141

Regarding the potential endogeneity issues in the model, firstly, there might be omitted variable bias and a reverse causality problem. On the one hand, in the regression, we cannot control for all factors that may influence enterprise carbon emission intensity. On the other hand, the carbon emission intensity of an enterprise may affect its decision to adopt industrial robots. To address potential omitted variable bias and reverse causation issues, in robustness tests, we employ two different instrumental variables to mitigate the impact of omitted variables and demonstrate that the baseline conclusions are robust against potential endogeneity issues.

Additionally, it is worth noting that our baseline model (1) is not strictly a difference-in-differences (DID) model since the ATS covers the years 2007-2016. Some enterprises might have imported robots before 2007. For such enterprises, we assume that they adopted robots for production activities throughout the sample period from 2007 to 2016 but do not consider 2007 as the specific year of robot adoption. Therefore, this model is not a strictly multi-time-point DID. Nevertheless, the model may still face a self-selection issue, meaning that companies adopting robots may have

had higher energy efficiency before adopting robots. To address this concern, in subsequent robustness tests, we use the Propensity Score Matching-DID (PSM-DID) method for regression to demonstrate that the conclusion of robot-induced carbon reduction remains robust. However, to maintain a rich sample in the baseline regression, we use data from the entire sample of manufacturing enterprises in the baseline regression.

5. Empirical results

5.1 Baseline results

The baseline regression results are presented in Table 2. It can be observed that as control variables and various fixed effects are included, the R-squared of the regression gradually increases, indicating an enhanced explanatory power of the model. Taking the results in column (4) as the baseline, the regression coefficient for $robot_{it}$ is -0.0606, suggesting that the utilization of industrial robots leads to a reduction in corporate carbon emission intensity by 6.06%.

Table 2: Baseline results

$ln ci_{it}$	(1)	(2)	(3)	(4)
$robot_{it}$	-0.1553*** (0.0130)	-0.1983*** (0.0131)	-0.0552** (0.0238)	-0.0606*** (0.0237)
C_{it}	No	Yes	Yes	Yes
Firm FE	No	No	Yes	Yes
Year×Industry FE	No	No	No	Yes
Observations	1390554	1390554	1091063	1090768
R ²	0.0001	0.0328	0.4042	0.4163

Notes: Firm-level clustered standard errors are reported in parentheses. *, **, and *** denote $p < 0.1$, $p < 0.05$, and $p < 0.01$, respectively. Unless otherwise specified, the same applies.

The reason may be that, following the adoption of robots, enterprises experience a reduction in energy waste and consumption during the production process. The energy required to produce one unit of goods decreases, enhancing the energy efficiency of the

enterprise and lowering carbon emission intensity. Simultaneously, the adoption of robots may also decrease labor costs for the company, encouraging further innovation in low-carbon technologies and subsequently reducing the carbon emission intensity of the enterprise. A more detailed analysis of the specific reasons for these effects is conducted in the subsequent Section 6.1 on mechanism testing.

5.2 Robustness tests

To demonstrate the robustness of our baseline regression results, we employ four different approaches to validate the stability of the conclusions.

5.2.1 Alternative measures of robot adoption

In the baseline regression, we use the indicator of whether an enterprise imports industrial robots as the core explanatory variable. In robustness tests, we replace this measurement with two alternative variables: the cumulative value of imported robots and the per capita cumulative value of imported robots (both logarithmic). The results are presented in columns (1) and (2) of Table 3, respectively. The regression coefficients remain significantly negative, affirming the robustness of our conclusion. Subsequently, we utilize industry-level robot stock data provided by IFR for various sectors in China, we match these two datasets based on the enterprise's industry affiliation and the corresponding year. The logarithm of the industrial robot stock in the sector to which the enterprise belongs is employed as a measure of the robot exposure rate (Liu et al., 2021). Due to the utilization of IFR data to measure robot adoption intensity at the industry-year level, controlling for industry-year fixed effects would result in complete collinearity, making it impossible to obtain regression coefficients for the core explanatory variables. Therefore, for the parts involving the measurement of robot intensity using IFR data, only quartile industry fixed effects and year fixed effects are controlled, without accounting for their interaction fixed effects, and the same applies subsequently. The results are shown in column (3) of Table 3, and the regression coefficients remain significantly negative, providing further support for the robustness of our conclusion.

5.2.2 Additional control variables and fixed effects

We introduce additional control variables and fixed effects to further validate the

robustness of the results. Specifically, we include enterprise age ($lnage_{it}$) and the debt ratio of the enterprise ($debratio_{it}$) in Model (1). Additionally, we introduce fixed effects for the city in which the enterprise is located. The results are displayed in column (4) of Table 3, and the regression coefficients for $robot_{it}$ remain significantly negative, confirming the robustness of the conclusion. However, this approach leads to a loss of observations, which is why these control variables and fixed effects were not introduced in the baseline regression.

5.2.3 Adjustment of clustering level

In the baseline regression, clustering is performed at the enterprise level. Due to significant industry heterogeneity in robot adoption (Liu et al., 2021), we increase the clustering level to the industry-year level. The results are shown in column (5) of Table 3. Although the significance level slightly decreases, the regression coefficients for $robot_{it}$ remain significantly negative at the 5% level, indicating the robustness of the conclusion.

5.2.4 Exclusion of the impact of carbon market pilot policies

In 2013, China initiated carbon emissions trading pilot programs in seven provinces and cities, affecting some high-carbon enterprises in those regions. To address this policy impact, we introduce a variable cbm_{it} , representing the impact of the carbon market pilot policy, assuming that enterprises in these seven regions were affected by the policy starting from 2013. The results, shown in column (6) of Table 3, reveal that the regression coefficient for the core explanatory variable remains significantly negative, confirming that the baseline results are not influenced by the carbon market pilot policy.

Overall, the robustness tests support and validate the stability of the conclusion that industrial robot usage reduces carbon emission intensity.

Table 3: Robustness testing results

$ln ci_{it}$	Adjust the robot measurement method	Increment	Adjust	Exclude
		control variable	the clustering	the influence

				and fixed effects	level	of carbon market
	(1)	(2)	(3)	(4)	(5)	(6)
$robot_{it}$	-0.0049*** (0.0019)	-0.0085** (0.0035)	-0.0045*** (0.0007)	-0.0590** (0.0255)	-0.0606** (0.0252)	-0.0630*** (0.0237)
C_{it}	Yes	Yes	Yes	Yes	Yes	Yes
cbm_{it}	No	No	No	No	No	0.0206*** (0.0033)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Year×Industry FE	Yes	Yes	Separate	Yes	Yes	Yes
Region FE	No	No	No	Yes	No	No
Observations	1090768	1090768	1091043	979982	1090768	1090768
R ²	0.4163	0.4163	0.4118	0.4156	0.4163	0.4163

Notes: Firm-level clustered standard errors are reported in parentheses of column (1)-(4) and column (6). Industry-Year level clustered standard errors are reported in the parentheses of column (5).

5.3 Endogeneity discussion

To demonstrate that our baseline regression results are robust to issues such as omitted variables, reverse causality, and self-selection bias, we employ instrumental variable and propensity score matching with difference-in-differences (PSM-DID) methods to address potential endogeneity in the model.

5.3.1 Instrumental variable method

We utilize the IV method to address potential issues related to omitted variables and reverse causality. The construction of the instrumental variable is as follows: (1) We use the logarithm of the number of industrial robots used by other firms in the same sector, excluding the focal firm itself, as the instrumental variable ($lnrobotnum_{it}$). (2) Leveraging IFR data, we employ the logarithm of the stock of industrial robots at the industry level in Portugal as the instrumental variable ($lnport_{it}$). This approach ensures

that the quantity of robots in other countries reflects only the technological characteristics of the industry, thereby avoiding endogeneity issues arising from domestic data. The second-stage regression results using these two instrumental variables are presented in Table 4, indicating a significantly negative impact, thus affirming that our baseline regression results are not affected by endogeneity issues. The first-stage regression results are displayed in Appendix Table A, with F-values exceeding 10, confirming the absence of weak instrument problems.

Table 4: IV results

$ln ci_{it}$	(1) $ln robotnum_{it}$	(2) $ln port_{it}$
$robot_{it}$	-0.0935*	-10.3219**
	(0.0506)	(4.1432)
C_{it}	Yes	Yes
Firm FE	Yes	Yes
Year×Industry FE	Yes	Separate
Observations	1090768	1091043
ROOT MSE	0.4346	0.4841

5.3.2 Reanalysis based on PSM-DID

According to Fan et al. (2021) and Huang et al. (2022), firms decide whether to adopt industrial robots based on their own production and operational conditions, leading to potential self-selection issues. To address this concern, we employ the PSM-DID method to select a more suitable control group for enterprises importing robots. Since the sample in ATS covers only the period from 2007 to 2016, and some firms engaged in robot imports before 2007, we restrict our analysis to samples of firms importing robots after 2007. To mitigate interference from this subset of samples in the regression results, we adopt a yearly PSM-DID approach, using nearest neighbor matching in both 1:1 and 1:4 without replacement, to select control groups for treatment groups. The point at which a firm first imports a robot during the period 2007-2016 is considered as the treatment point. The calculation of propensity scores is illustrated in Equation (2):

$$Pr(robot_{it} = 1) = \Phi\{h(C_{i,t-1}, \ln CI_{i,t-1})\} \quad (2)$$

In this analysis, lagged one-period values are employed for matching variables. The matching variables $C_{i,t-1}$, include firm size $\ln size_{i,t-1}$, capital intensity $\ln kratio_{it}$, profitability $\ln profit_{it}$, human capital level $\ln hc_{it}$, and carbon emission intensity $\ln ci_{i,t-1}$. The inter-group differences before and after matching are presented in Appendix Table B, indicating that propensity score matching has mitigated differences between the treatment and control groups to a certain extent. The results of PSM-DID are displayed in Table 5, where the core explanatory variable remains significantly negative. This demonstrates that our baseline regression results are not affected by self-selection issues. Furthermore, parallel trends tests have been conducted, and the results are presented in Appendix Table C, revealing no significant differences in carbon emission intensity between the treatment and control groups before the adoption of industrial robots. This provides evidence that the reduction in carbon emission intensity is indeed attributable to the adoption of industrial robots.

Table 5: PSM-DID results

$\ln ci_{it}$	(1) 1: 1	(1) 1: 4
$robot_{it}$	-0.0767** (0.0364)	-0.0696** (0.0327)
C_{it}	Yes	Yes
Firm FE	Yes	Yes
Year×Industry FE	Yes	Yes
Observations	4135	9775
R ²	0.2306	0.2953

6. Mechanism and heterogeneity analysis

6.1 Mechanism examination

6.1.1 Productivity enhancement effect

In this section, we introduce the total factor productivity obtained from the Olley-Pakes method (tfp_op_{it}) and the Levinsohn-Petrin method (tfp_lp_{it}) as the dependent

variables in Equation (1). The regression results are shown in columns (1) and (2) of Table 6. A positive and significant regression result would indicate that the adoption of robots indeed improves the overall firm productivity, thus reduce the enterprises' carbon emission intensity. As discussed in the theoretical analysis earlier, the adoption of robots can achieve uninterrupted production 24 hours a day, executing repetitive and tedious tasks at higher speeds and consistent quality, which can significantly enhance firm productivity, enabling firms to achieve equivalent output with less energy input in the production process. Consequently, firms experience increased energy efficiency, leading to a natural reduction in carbon emission intensity. Thus, hypothesis 1 is validated.

6.1.2 Labor cost saving effect

We measure firm labor intensity ($lnten_{it}$) using the ratio of the average number of employees to operating income and firm labor cost ($lcost_{it}$) using the ratio of total employee compensation to operating income, these indicators can be used to represent the labor costs of enterprises. After taking logarithms, these variables are introduced as dependent variables in Equation (1). The regression results in columns (3) and (4) of Table 6 demonstrate a significantly negative relationship, indicating that the substitution of low-skilled labor through the adoption of robots enables cost savings in labor expenses. These saved costs can then be redirected towards purchasing more clean energy or clean equipment, thereby reducing carbon emissions at the source. Alternatively, enterprises may invest in carbon offset projects to counterbalance their carbon emissions. Thus, Hypothesis 2 is validated.

6.1.3 Low-carbon technological innovation effect

To test whether industrial robot adoption indeed brings about more low-carbon technologies, we merge ATS data with the Chinese National Patent Database based on firm names and years. We classify patents into green low-carbon patents based on the patent classification system outlined in the "Green Low-Carbon Technology Patent Classification System" published by the China National Intellectual Property Administration. We sum up the number of new green low-carbon patents applied for by each firm-year and introduce the logarithmically transformed variable ($lnlct_{it}$) as the

dependent variable in Equation (1). The regression result in column (5) of Table 6 reveals a significantly positive coefficient, indicating that the adoption of robots can stimulate firms to engage in low-carbon technological innovation. The reason may be that adopting advanced production technologies, such as industrial robots, may involve novel and unique methods and systems, fostering low-carbon technological innovations. In addition, the substitution of low-skilled labor by robots promotes an upgrade in the human capital structure. This shift makes enterprises place greater emphasis on innovation and low-carbon production, thereby fostering innovation in low-carbon technologies within the enterprise. Furthermore, the development of more advanced low-carbon technologies leads to a subsequent reduction in carbon emission intensity. Thus, Hypothesis 3 is validated.

Table 6: Mechanism test results

	Productivity enhancement		Labor cost saving		Low-carbon technology innovation
	(1)	(2)	(3)	(4)	(5)
	tfp_op_{it}	tfp_lp_{it}	$lnlinten_{it}$	$lnlcost_{it}$	$lnlct_{it}$
$robot_{it}$	0.3397*** (0.0803)	0.3398*** (0.0803)	-0.0246** (0.0103)	-0.1322*** (0.0289)	0.0419** (0.0192)
C_{it}	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes
Year×Industry FE	Yes	Yes	Yes	Yes	Yes
Observations	782234	782234	984066	984065	1090768
R ²	0.3122	0.2944	0.1847	0.3275	0.4133

6.2 Heterogeneity analysis

6.2.1 High-carbon firms

Firstly, we examine whether the carbon emission intensity of firms affects the emission reduction effect of adopting robots. To investigate this, we generate a dummy

variable $cbdum_{it}$ representing whether a firm's carbon emission intensity is above the median of the entire sample. If the carbon emission intensity is higher than the median, the dummy variable takes a value of 1; otherwise, it takes a value of 0. We create an interaction term between this dummy variable and $robot_{it}$, and introduce both into the regression model (1). The results in column (1) of Table 7 show that the coefficient of the interaction term is significantly negative. This indicates that high-carbon firms experience a greater reduction in carbon emission intensity after adopting robots, facilitating their low-carbon transformation and contributing to the achievement of China's dual carbon goals.

6.2.2 Firm size

Furthermore, we investigate whether firm size influences the reduction in carbon emission intensity through the adoption of robots. Specifically, we divide firms into relatively large and small scale enterprises based on the median number of employees in the entire sample. We create an interaction term between this classification ($sizedum_{it}$) and $robot_{it}$, and introduce it into the baseline model. The results in column (2) of Table 7 demonstrate that the coefficient of the interaction term is significantly negative. This suggests that larger-scale enterprises experience a more pronounced reduction in carbon emission intensity after adopting robots. Larger firms typically emit higher amounts of carbon, and the reduction in national-level carbon emissions is primarily contributed by the decarbonization efforts of industrial giants (Guo et al., 2020). This finding supports the notion that industrial robots are aiding larger-scale enterprises in achieving a more effective low-carbon transformation.

6.2.3 Export status

Lastly, we further categorize firms into two groups based on whether they engage in export activities. We anticipate that export-oriented firms generally have higher productivity and face more stringent environmental standards for their exported products. Thus, export-oriented firms may already exhibit better environmental performance prior to adopting robots. Indeed, in our regression sample, the average carbon emission intensity for export-oriented firms is 0.18, compared to 0.27 for non-exporting firms. We introduce an interaction term between the export status $exdum_{it}$

and $robot_{it}$ into the regression model. The results in column (3) of Table 7 show that the coefficient of the interaction term is significantly positive. This confirms our hypothesis that non-exporting firms experience a greater reduction in carbon emission intensity after adopting robots. This result suggests that for non-exporting firms or those with relatively poorer performance, the adoption of industrial robots is especially beneficial, not only in improving productivity but also in enhancing environmental performance, facilitating a successful low-carbon transformation.

Table 7: Heterogeneity analysis

$ln ci_{it}$	(1)	(2)	(3)
$robot_{it}$	0.0148 (0.0248)	-0.0022 (0.0284)	-0.0956*** (0.0342)
$robot_{it} * cb dum_{it}$	-0.1230*** (0.0258)	No	No
$robot_{it} * sizedum_{it}$	No	-0.0612** (0.0288)	No
$robot_{it} * ex dum_{it}$	No	No	0.0555** (0.0278)
Characteristic dummy	Yes	Yes	Yes
C_{it}	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes
Year×Industry FE	Yes	Yes	Yes
Observations	1090768	1090768	1090768
R ²	0.4437	0.4163	0.4163

7. Further analysis

In the previous baseline regression, we conducted an analysis of enterprise carbon intensity. In this section, we begin by analyzing total carbon emissions of enterprises and discuss the impact of industrial robot adoption on overall carbon emissions. Subsequently, we further integrate information on exported goods provided by customs

data to examine whether robot adoption has altered the range of goods produced by enterprises, thereby reducing corporate carbon intensity. Finally, we examine whether the emission reduction effect of robot adoption has caused carbon leakage.

7.1 Total carbon emissions and total output

In the preceding research, we focused on the carbon intensity of enterprise unit output. In this section, our focus shifts to total corporate carbon emissions ($Intce_{it}$), observing its susceptibility to the influence of industrial robot adoption. Specifically, we employ the logarithm of total enterprise carbon emissions as the dependent variable in regression model (1), the results of which are displayed in Table 8, Column (1). The coefficient of the core explanatory variable in the regression is significantly positive, demonstrating that although the carbon intensity per unit output decreases after the adoption of robots, overall carbon emissions increase. As total carbon emissions equal the product of carbon emissions per unit output and total output, the regression results for total output ($lnoutput_{it}$) are presented in Table 8, Column (2). The increase in total carbon emissions, coupled with the simultaneous decrease in carbon intensity and rise in total output, indicates that the expansive effect of robot adoption on total output is more substantial. This results in a decline in carbon intensity that cannot offset the increased carbon emissions from heightened production, ultimately leading to an overall upward trend in carbon emissions.

However, we observe variations in the extent to which total output is influenced by robot adoption, depending on the carbon intensity of individual enterprises. Utilizing the methodology to measure high-carbon enterprises from the heterogeneity analysis in the previous section, we generate an interaction term between robot adoption and high-carbon enterprises ($cbdum_{it}$). Regression results for total carbon emissions and total output are presented in Table 8, Column (3). The significant negative coefficient of the interaction term indicates that enterprises with higher carbon intensity experience a relatively lower increase in total output after the adoption of robots. Combining this result with the findings from the heterogeneity analysis in Section 5.2.1, where high-carbon enterprises exhibited a greater reduction in carbon intensity after the adoption of robots, it is essential to clarify that the conclusion that robot adoption leads to a

greater decrease in carbon intensity for high-carbon enterprises is not attributed to a larger increase in total output. Instead, it is primarily due to a substantial reduction in energy consumption facilitated by the adoption of robots. This reduction in energy consumption to a certain extent alleviates the environmental burden of corporate production activities, contributing to the transition of enterprises and China towards environmentally friendly low-carbon production.

Table 8: Robot adoption and enterprises' total carbon emissions

	(1) $lnfce_{it}$	(2) $lnoutput_{it}$	(3) $lnoutput_{it}$
$robot_{it}$	0.4165*** (0.1346)	0.3031*** (0.0610)	0.4167*** (0.0773)
$robot_{it} * cbdum_{it}$	No	No	-0.1934*** (0.0725)
$cbdum_{it}$	No	No	Yes
C_{it}	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes
Year×Industry FE	Yes	Yes	Yes
Observations	1090768	1090715	1090715
R ²	0.5975	0.8170	0.8170

7.2 Adjustment in the range of produced goods by enterprises

Subsequently, we further examine whether the adoption of robots leads to adjustments in the structure of goods produced by enterprises. Specifically, different products generate varying levels of carbon emissions during the production process. We aim to investigate whether robot adoption prompts enterprises to produce a higher proportion of low-carbon products, thereby facilitating the transition towards low-carbon practices. Since we do not have access to information about the specific goods produced by enterprises, and export data from manufacturing companies can to some extent reflect the production structure, we merge ATS with CCD to observe changes in the range of goods produced by enterprises before and after the adoption of robots. Concerning the measurement of carbon emissions generated during the production

process of each exported product, the World Input-Output Database (WIOD) provides data on the total carbon emissions and total output produced by various sectors in different countries. This enables us to obtain the carbon intensity of the production process for each sector. Subsequently, by unifying the Harmonized System (HS) codes of customs goods and matching them with the WIOD industry classification¹, we can derive the carbon emission structure intensity for each product in each year. The specific calculation method is as follows:

$$ci_str_{it} = \sum_{j \in J} carbon_{jt} * \frac{exvalue_{ijt}}{\sum_{j \in J} exvalue_{ijt}} \quad (3)$$

Where the subscript j represents the j -th HS8 code of goods exported by enterprise i in year t , $j \in J$ is the set of all HS8 code goods exported by enterprise i in year t , $carbon_{jt}$ denotes the carbon emission intensity per unit output for the j -th product in the production process in year t , and $exvalue_{ijt}$ represents the amount of product j exported by enterprise i in year t . The variable ci_str_{it} , calculated using Equation (3), reflects the carbon emission structure of the goods produced by exporting enterprises. After taking the logarithm of ci_str_{it} , it is introduced as the dependent variable in regression model (1). The regression results are presented in Table 9, Column (1), where the coefficient of the core explanatory variable is not statistically significant. This suggests that, overall, the adoption of robots does not significantly alter the structure of goods produced by enterprises. However, we hypothesize that if enterprises actively engage in low-carbon transformation after adopting robots, the structure of goods they produce may tend towards lower carbon emission intensity.

To validate this hypothesis, we use the variable indicating whether the enterprise has applied for green and low-carbon patents ($lctdum_{it}$) as a measure of the enterprise's low-carbon transformation. This choice is based on the notion that the application for green and low-carbon patents reflects the efforts and investments made by enterprises in reducing their own carbon emissions. Subsequently, we generate the interaction term

¹ Due to the limitations of the World Input-Output Database (WIOD), which only provides the total carbon emissions for various industries in China before 2014, and given that we observed minimal changes in industry carbon emission intensity from 2007 to 2014, we opt to substitute the carbon emission intensity for the years 2015 and 2016 with the values from 2014.

between robot adoption and this variable, and employ model (1) to regress the carbon emission structure of goods produced by enterprises (ci_str_{it}). The regression results are presented in Table 9, Column (2), where the coefficient of the interaction term is significantly negative. This indicates that when enterprises actively engage in low-carbon transformation, the structure of goods they produce tends towards a lower carbon emission intensity. This not only contributes to enterprises effectively reducing their own carbon intensity and fulfilling their corporate responsibility towards low-carbon practices but also facilitates the overall national transition towards low-carbon production, leading to carbon neutrality.

Table 9: Robot adoption and enterprises' product range adjustment

$inci_str_{it}$	(1)	(2)
$robot_{it}$	-0.0046 (0.0603)	0.0199 (0.0610)
$robot_{it}*lctdum_{it}$	No	-0.1828** (0.0858)
$lctdum_{it}$	No	Yes
C_{it}	Yes	Yes
Firm FE	Yes	Yes
Year×Industry FE	Yes	Yes
Observations	194179	194179
R ²	0.7635	0.7635

7.3 Is there carbon leakage?

Finally, we further examine whether the decline of carbon emissions in the manufacturing sector due to the adoption of robots may result in carbon leakage. Specifically, while the adoption of robots reduces enterprises' consumption of fossil fuels and directly decreases carbon emissions from manufacturing, it may also lead to an increase in electricity consumption. The rise in electricity demand from the manufacturing sector may cause manufacturing enterprises to shift decarbonization pressure to the electricity sector, potentially leading to carbon leakage issues. To

investigate this concern, we calculate the electricity consumption per unit of output ($lnele_{it}$) and introduce it as the explanatory variable into model (1) for regression analysis. The regression results are presented in Table 10.

Table 10: Carbon Leakage

$lnele_{it}$	(1)
$robot_{it}$	0.0089 (0.0100)
C_{it}	Yes
Firm FE	Yes
Year×Industry FE	Yes
Observations	1032946
R^2	0.0809

It can be observed that the regression coefficient of the core explanatory variable is not significant, indicating that the adoption of robots does not increase the intensity of electricity usage in enterprises. This result suggests to a certain extent that the emission reduction effect of using robots in manufacturing enterprises does not come at the expense of increased decarbonization pressure on the electricity sector. Therefore, there is no carbon leakage issue. This is beneficial for the overall reduction of carbon intensity in China and the achievement of dual carbon goals.

8. Conclusion

This study utilized Chinese manufacturing enterprise data from the ATS for the years 2007 to 2016, coupled with information on robot imports obtained from CCD, to investigate whether the adoption of robots contributes to a reduction in carbon emissions for enterprises. The findings reveal that the adoption of industrial robots significantly decreases the carbon intensity of manufacturing industry enterprises, this effect is more pronounced in high-carbon enterprises, larger-scale enterprises, and non-exporting enterprises. The mechanism analysis elucidates that productivity enhancement, labor cost saving and low-carbon technological innovation are the main

channels. Further analysis reveals that while the adoption of robots to some extent lowers the carbon emissions intensity per unit output, the overall carbon emissions of enterprises show an increasing trend due to a greater increase in total output. Nonetheless, a relatively lower increase in output is observed for high-carbon enterprises. Subsequent analysis employs export product information from the CCD to measure the range of goods produced by enterprises. We innovatively calculated the trade-weighted carbon emission structure intensity of goods produced by enterprises and found that the adoption of robots significantly promotes the reduction of carbon emission structure intensity when enterprises invest more effort in green and low-carbon transformation. Finally, we found that the adoption of robots did not increase the electricity usage intensity per unit of output in enterprises, demonstrating that the reduction in carbon emissions intensity in the manufacturing sector did not come at the expense of increased electricity generation in the electricity sector, and thus, did not result in carbon leakage. Based on these research conclusions, the following policy recommendations are proposed:

Firstly, manufacturing enterprises can choose to adopt robots based on their actual production conditions and operational status. The adoption of robots not only drives the low-carbon transformation of enterprises but also enhances their productivity. Intelligent manufacturing contributes to a win-win situation for both the economy and the environment. Secondly, despite the cost savings in labor and enhanced production efficiency resulting from the adoption of robots, precautions should be taken against the potential risk of mass unemployment due to "machine substitution". In addition, enterprises aiming for a transition to intelligent manufacturing should actively engage in employee skills training and hire more high-skilled talents to meet the demands of intelligent production. Thirdly, high-carbon emitting enterprises and large-scale enterprises should consider the adoption of robots as a means to reduce their own carbon emissions intensity. These enterprises play a crucial role in the national economy and are significant contributors to emission reduction tasks, serving as the primary force for achieving dual carbon goals. Given that robot adoption can effectively help these enterprises reduce carbon emissions intensity, relevant government

departments can alleviate financial pressures on enterprises purchasing and adopting robots through policies such as tax exemptions and R&D funding. Finally, although the adoption of robots can reduce the carbon emissions intensity of enterprises, measures should be taken to avoid the disorderly expansion of production output leading to an increased environmental burden. Additionally, increased investment in green and low-carbon technological innovation is recommended. With the introduction of carbon peaking and carbon neutrality goals, enterprises face higher carbon reduction requirements. Merely reducing carbon emissions intensity may not be sufficient to meet China's current carbon neutrality needs. Therefore, the government should implement more robust green monetary policies and financial subsidies, encouraging enterprises to allocate more resources to innovation in green and low-carbon technologies. This involves enhancing energy efficiency at the source, incorporating more clean energy, and adopting advanced carbon capture technologies to effectively reduce overall carbon emissions.

Appendix

Firstly, we present the first stage regression results of the instrumental variable method in Section 5.3.1 in Appendix Table A.

Appendix Table A: The first-stage regression results for the instrumental variables.

$robot_{it}$	(1)	(2)
$lnrobotnum_{it}$	-1.3141*** (0.0523)	No
$lnport_{it}$	No	0.0006*** (0.0001)
C_{it}	Yes	Yes
Firm FE	Yes	Yes
Year×Industry FE	Yes	Separate
Observations	1090768	1091043

F value	630.98	20.39
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Next, we display the pre-treatment imbalances in covariates between the treatment and control groups before and after propensity score matching, the results for 1:1 and 1:4 matching are presented in Appendix Table B.

Appendix Table B1: Propensity score matches results of 1v1 matching

Variable	Unmatched	Mean		bias%
	Matched	Treated	Control	
$ln ci_{it}$	U	0.1297	0.2561	97.46%
	M	0.1297	0.1469	13.26%
$ln size_{it}$	U	6.3480	4.1963	-33.90%
	M	6.3480	6.4311	1.310%
$ln klratio_{it}$	U	5.0463	3.8712	-23.29%
	M	5.0463	4.8260	-4.37%
$ln hc_{it}$	U	3.6573	3.0586	-16.37%
	M	3.6573	3.4797	-4.86%
$ln profit_{it}$	U	6.6348	1.7780	-73.20%
	M	6.6348	6.9442	4.660%
$state_{it}$	U	0.0107	0.0482	350.5%
	M	0.0107	0.0194	81.31%

Appendix Table B2: Propensity score matches results of 1v4 matching

Variable	Unmatched	Mean		bias%
	Matched	Treated	Control	
$ln ci_{it}$	U	0.1297	0.2561	97.46%
	M	0.1297	0.1715	32.23%
$ln size_{it}$	U	6.3480	4.1963	-33.90%
	M	6.3480	6.2536	-1.487%
$ln klratio_{it}$	U	5.0463	3.8712	-23.29%

	M	5.0463	4.9489	-1.930%
$lnhc_{it}$	U	3.6573	3.0586	-16.37%
	M	3.6573	3.4969	-4.386%
$lnprofit_{it}$	U	6.6348	1.7780	-73.20%
	M	6.6348	6.1432	-7.409%
$state_{it}$	U	0.0107	0.0482	350.5%
	M	0.0107	0.0221	106.54%

We adopted the following model to examine the parallel trends, where j represents the number of years since the first industrial robot import by the enterprise after 2007. For instance, when $j=1$, it signifies the year immediately following the enterprise's first import of industrial robots. We conducted parallel trends tests for both 1:1 matching and 1:4 matching, with $j=-1$ selected as the control group. Regression results are presented in Appendix Table C, revealing no significant differences before the adoption of industrial robots regardless of the matching method used, as confirmed by the parallel trends test.

$$lni_{it} = \sum_{j=-8}^7 \beta_j Post_j \times robot_i + \varepsilon_{it}$$

Appendix Table C: Parallel trends test

lni_{it}	(1) 1-1 matching	(3) 1-4 matching
$Post_{<=-4} \times robot_i$	0.0107 (0.0536)	-0.0007 (0.0494)
$Post_{-3} \times robot_i$	-0.0529 (0.0383)	-0.0547 (0.0393)
$Post_{-2} \times robot_i$	0.0437 (0.0439)	0.0510 (0.0569)
$Post_0 \times robot_i$	-0.0634 (0.0418)	-0.0538 (0.0421)
$Post_1 \times robot_i$	-0.0669	-0.0601

	(0.0457)	(0.0444)
$Post_2 \times robot_i$	-0.0766	-0.0791
	(0.0496)	(0.0487)
$Post_3 \times robot_i$	-0.1337**	-0.1085**
	(0.0574)	(0.0541)
$Post_{\geq 4} \times robot_i$	-0.0572	-0.0381
	(0.0967)	(0.0767)
Firm FE	Yes	Yes
Year×Industry FE	Yes	Yes
Observations	4135	9775
R ²	0.2306	0.2954

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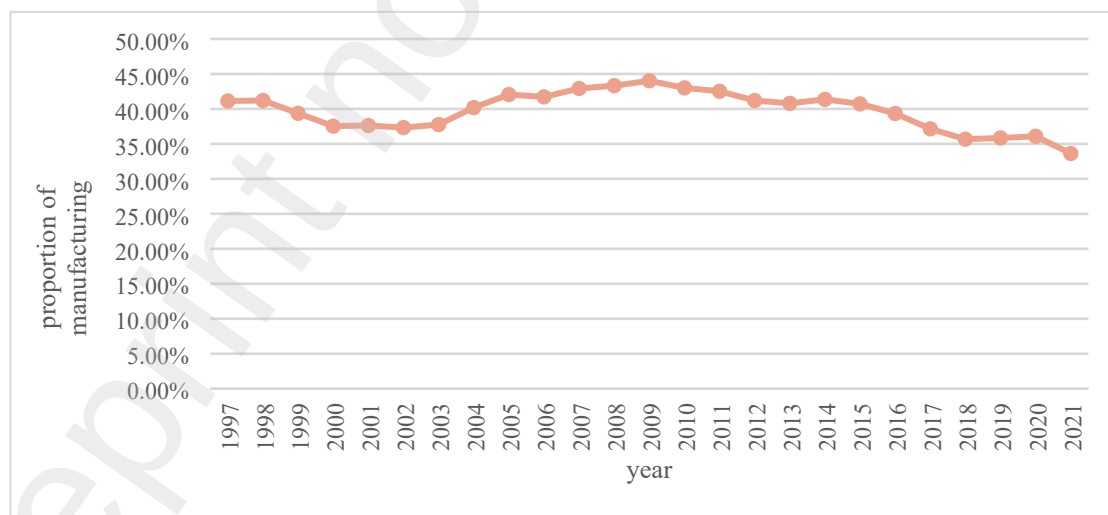


Fig 1. Trend chart of carbon emission proportion of manufacturing industry in China